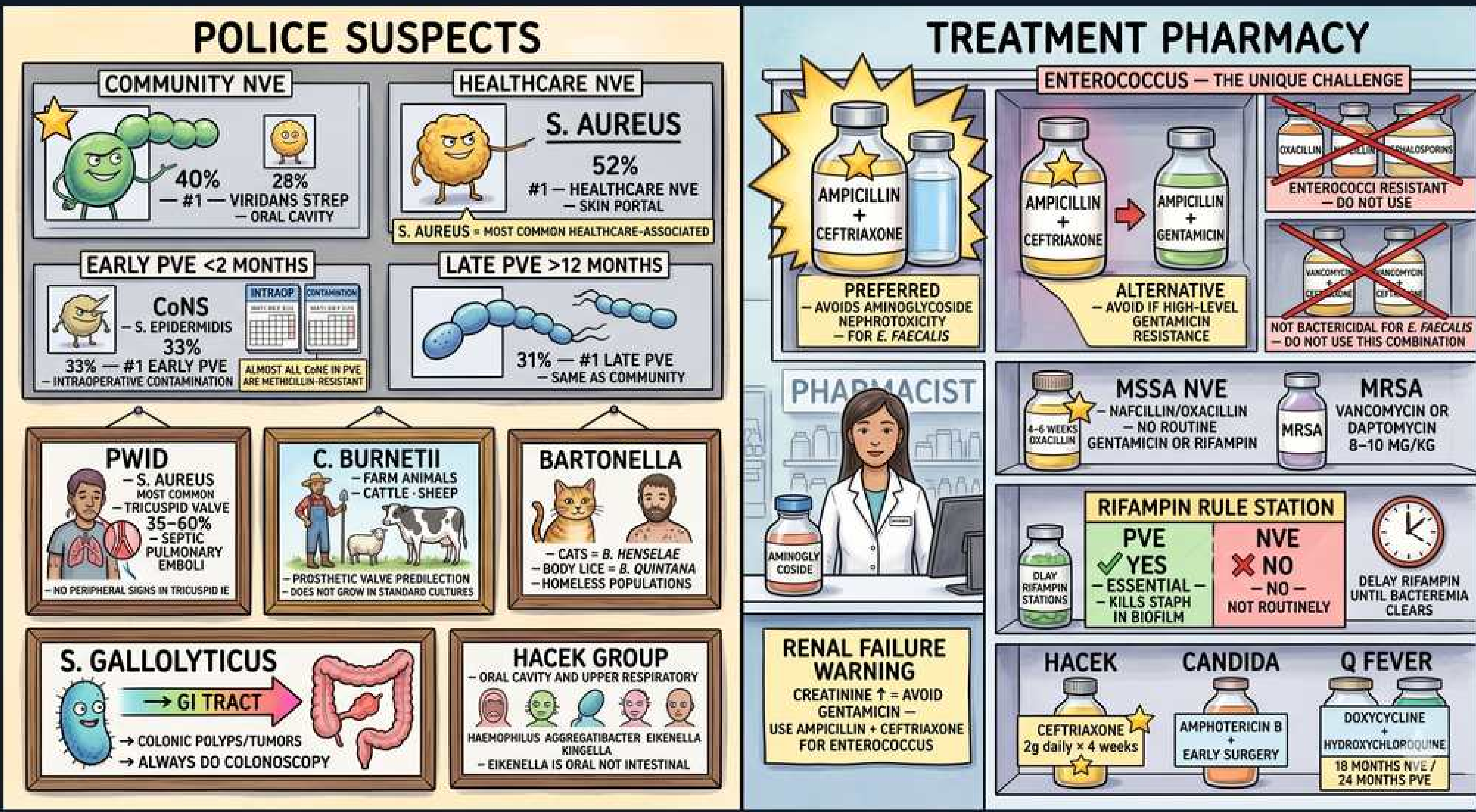
Infective Endocarditis — Microbiology & Organism-Specific Therapy



1. Community NVE — Viridans

WHAT YOU SEE

Community NVE: ~40% Viridans strep, oral cavity

WHAT IT ENCODES

In community-acquired native-valve IE, Viridans streptococci (oral flora, ~40%) are the classic cause, typically subacute. Treated with penicillin/ceftriaxone +/- gentamicin for synergy.

BOARD TRAP

Subacute, oral-source IE points to Viridans strep — not S. aureus, which is acute/healthcare-associated.

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2. S. aureus — overall #1

WHAT YOU SEE

~28% S. aureus community / 52% healthcare

WHAT IT ENCODES

S. aureus is the single most common IE organism overall and predominates in healthcare-associated NVE (~52%) and IVDU. Causes acute, destructive, often right-sided (IVDU) disease.

BOARD TRAP

S. aureus, not strep, is the #1 overall and the #1 healthcare/IVDU cause — acute presentation.

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3. CoNS / early PVE

WHAT YOU SEE

Early PVE <2 months: CoNS, S. epidermidis 33%

WHAT IT ENCODES

Coagulase-negative staph (S. epidermidis) is the leading cause of early prosthetic-valve endocarditis (<2 months), reflecting perioperative seeding. Often methicillin-resistant → vancomycin-based.

BOARD TRAP

Early PVE = CoNS (perioperative); late PVE (>12 months) resembles community organisms like Viridans strep.

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4. Late PVE resembles community

WHAT YOU SEE

Late PVE >12 months: ~31% same as community

WHAT IT ENCODES

Late PVE (>12 months) microbiology mirrors community NVE (Viridans strep, S. aureus, enterococcus) because it represents new community exposure rather than surgical contamination.

BOARD TRAP

Don't assume all PVE is staph — late PVE looks like native community IE.

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5. IVDU — tricuspid S. aureus

WHAT YOU SEE

PWID: S. aureus, tricuspid valve, septic emboli

WHAT IT ENCODES

People who inject drugs get right-sided (tricuspid) IE, usually S. aureus, presenting with septic pulmonary emboli rather than peripheral arterial emboli; often no peripheral stigmata.

BOARD TRAP

Right-sided IVDU IE causes SEPTIC PULMONARY emboli (lung nodules), not Osler/Janeway peripheral stigmata.

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6. S. gallolyticus → colonoscopy

WHAT YOU SEE

S. gallolyticus / GI tract / colonoscopy

WHAT IT ENCODES

Streptococcus gallolyticus (formerly S. bovis) IE is strongly associated with colonic polyps/carcinoma — every case mandates colonoscopy to find an occult GI malignancy.

BOARD TRAP

S. bovis/gallolyticus IE = mandatory colonoscopy; missing the colon cancer link is a classic exam error.

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7. Coxiella / Bartonella culture-neg

WHAT YOU SEE

C. burnetii (farm/cattle), Bartonella (cats/homeless)

WHAT IT ENCODES

Culture-negative IE: Coxiella burnetii (Q fever — livestock exposure, prosthetic valves) and Bartonella (cats=henselae; homeless/lice=quintana). Diagnosed serologically, not by routine culture.

BOARD TRAP

Negative blood cultures don't exclude IE — pursue Coxiella/Bartonella/HACEK serologies.

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8. HACEK group

WHAT YOU SEE

HACEK: oral/upper respiratory flora

WHAT IT ENCODES

HACEK organisms (Haemophilus, Aggregatibacter, Cardiobacterium, Eikenella, Kingella) are oral/respiratory gram-negatives causing culture-negative-appearing IE; treat with ceftriaxone.

BOARD TRAP

HACEK IE is treated with ceftriaxone, not vancomycin — gram-negative oral flora.

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9. Enterococcus challenge

WHAT YOU SEE

Enterococcus — the unique challenge banner

WHAT IT ENCODES

Enterococcal IE needs cell-wall agent PLUS synergy: ampicillin + ceftriaxone (avoids aminoglycoside nephrotoxicity) is preferred; ampicillin + gentamicin is alternative if not high-level resistant.

BOARD TRAP

Ampicillin/penicillin ALONE is not bactericidal for enterococcus — needs synergistic dual therapy.

10. Ampicillin + ceftriaxone

WHAT YOU SEE

PREFERRED: ampicillin + ceftriaxone, avoids aminoglycoside

WHAT IT ENCODES

Ampicillin + ceftriaxone is preferred enterococcal regimen, especially with renal impairment or high-level aminoglycoside resistance, because it avoids gentamicin nephrotoxicity.

BOARD TRAP

With renal impairment or HLAR enterococcus, choose amp+ceftriaxone — not amp+gentamicin.

11. MSSA — nafcillin

WHAT YOU SEE

MSSA: nafcillin/oxacillin, +gentamicin/rifampin not routine

WHAT IT ENCODES

Methicillin-susceptible S. aureus NVE: nafcillin or oxacillin (cefazolin if allergic). Routine addition of gentamicin or rifampin is NOT recommended in native-valve disease.

BOARD TRAP

Don't add gentamicin/rifampin to MSSA native-valve IE routinely — increases toxicity without benefit.

12. MRSA — vancomycin/dapto

WHAT YOU SEE

MRSA: vancomycin or daptomycin 8-10 mg/kg

WHAT IT ENCODES

MRSA IE: vancomycin or daptomycin (8-10 mg/kg, higher dosing for bacteremia/IE). Daptomycin is inactivated by surfactant so it is for endovascular, not pulmonary, infection.

BOARD TRAP

Daptomycin is NOT used for pneumonia (surfactant inactivation) — but is fine for MRSA bacteremia/right-sided IE.

13. Rifampin — prosthetic only

WHAT YOU SEE

Rifampin rule station: PVE yes, NVE not routine

WHAT IT ENCODES

Rifampin is added for PROSTHETIC-valve staph IE (biofilm penetration on hardware) — start it after bacteremia clears. It is not used routinely in native-valve staph IE.

BOARD TRAP

Rifampin belongs to prosthetic-valve staph IE; adding it to native-valve disease is wrong and risks resistance.

14. Renal failure / gent avoidance

WHAT YOU SEE

Renal failure warning: avoid gentamicin

WHAT IT ENCODES

In renal impairment, avoid aminoglycoside-containing regimens; use ceftriaxone-based combinations (e.g., ampicillin + ceftriaxone 4 weeks for enterococcus) to spare the kidney.

BOARD TRAP

Defaulting to gentamicin in a patient with poor renal function is the wrong move — choose ceftriaxone synergy.

15. Q fever — doxy+HCQ long-term

WHAT YOU SEE

Q fever: doxycycline + hydroxychloroquine 18-24 months

WHAT IT ENCODES

Chronic Q fever (Coxiella) IE requires prolonged doxycycline + hydroxychloroquine for 18-24 months with serologic monitoring; HCQ alkalinizes phagolysosome to make doxy bactericidal.

BOARD TRAP

Short antibiotic courses fail Coxiella IE — it needs 18-24 months of doxycycline + hydroxychloroquine.